



Department Department of Electrical & Electronic Engineering

Rajshahi University of Engineering & Technology

LAB Report

Course Code : **EEE-3100**

Course Title : **Electronics Shop Practice**

Experiment No. : 07

Experiment Name : Implementation of Bluetooth and Voice-Controlled Home Automation System using HC-05 and Arduino.

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Experiment No.: 07

Experiment Name: Implementation of Bluetooth and Voice-Controlled Home Automation System using HC-05 and Arduino.

Objective:

- To understand the working and interfacing of the HC-05 Bluetooth module with Arduino.
- To develop a Bluetooth-based communication link between a smartphone and Arduino.
- To implement voice-controlled commands for operating home appliances.
- To control electrical devices (such as LEDs or relays) using Arduino based on received commands.
- To design a low-cost, scalable home automation system that can be operated wirelessly.
- To demonstrate the integration of mobile app and Arduino for remote control of home appliances.

Theory:

Home automation refers to the use of technology to control household appliances and systems such as lighting, fans, and security devices, making homes more convenient, efficient, and secure. This experiment involves the implementation of a Bluetooth and voice-controlled home automation system using an Arduino Uno and an HC-05 Bluetooth module.

The HC-05 Bluetooth module is a key component in this system. It operates on a DC voltage between 3.6V and 6V and communicates with microcontrollers through a UART (Universal Asynchronous Receiver/Transmitter) serial interface. The module is equipped with an integrated antenna, which enhances the stability and efficiency of wireless communication, typically within a range of about 10 meters (33 feet). It supports both master and slave modes and can be configured using AT commands to change settings such as device name, baud rate, and operating mode.

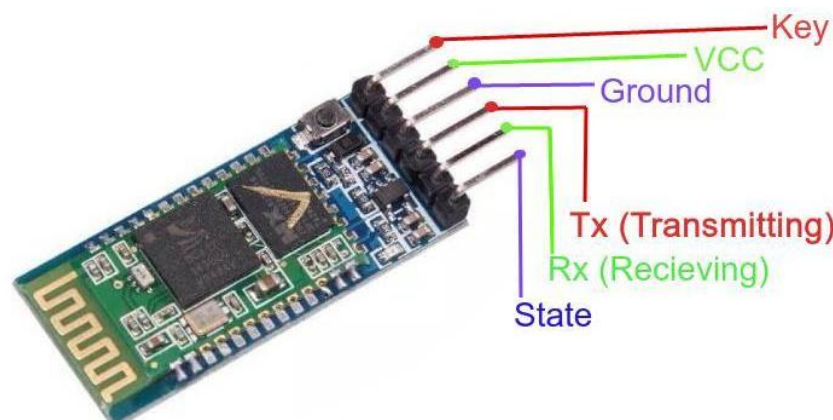


Fig.07.1: HC-05 Bluetooth Module

In this system, the HC-05 is connected to the Arduino using its TX and RX pins, allowing it to receive commands wirelessly from a Bluetooth-enabled smartphone. These commands can be either typed in a Bluetooth terminal app or generated by voice commands through a mobile app.

with voice recognition capability. The Arduino processes the received commands and controls connected appliances accordingly.

The electrical devices are controlled using a relay module, such as a 2-channel relay. A relay is an electrically operated switch that allows the Arduino to control high-voltage devices like lights or fans. It works using an electromagnetic mechanism and provides isolation between the low-voltage control circuit (Arduino) and the high-voltage load, ensuring safe operation.

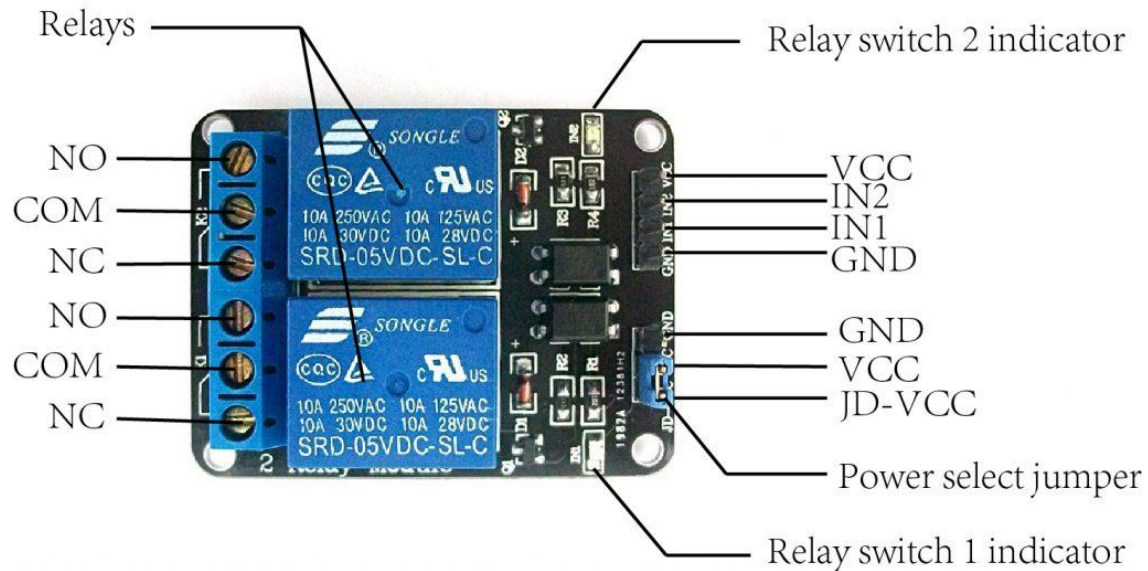


Fig.07.2: 2-Channel Relay Module

By integrating Bluetooth communication with voice control, this project demonstrates a practical, cost-effective solution for wireless home automation. The system enhances user convenience and is particularly beneficial for elderly or differently-abled individuals who may face difficulty operating traditional switches.

Required Apparatus:**Table 7.1: Table For Required Apparatus**

Sl. No.	Components	Specification	Quantity
1.	Arduino Uno	• ATmega328P • Operating Voltage: 5 V • Input Voltage: 7–12 V • Input Voltage: 6–20 V • Digital I/O Pins: 14 • PWM Digital I/O Pins: 6 (Pins 3, 5, 6, 9, 10, 11) • Analog Input Pins: 6 (A0 to A5) • DC Current per I/O Pin: 20 mA • DC Current for 3.3 V Pin: 50 mA • Flash Memory: 32 KB (ATmega328P) of which 0.5 KB used by bootloader • SRAM: 2 KB (ATmega328P) • EEPROM: 1 KB (ATmega328P) • Clock Speed: 16 MHz • USB Connection: Type B • LED Built-in: Pin 13	1
2.	Relay Module	• 2 Channel • Logic Input: 3.3-5V	1
3.	Bluetooth Module	• HC-05 • Operating Voltage: 3.6-6V DC	1
4.	AC Power	• 230 V RMS • 50 Hz	-
5.	Jumper Wires	-	-
6.	IDE Software	• Arduino IDE	-
7.	Light Bulb	• AC Type • 60W	

Circuit Diagram:

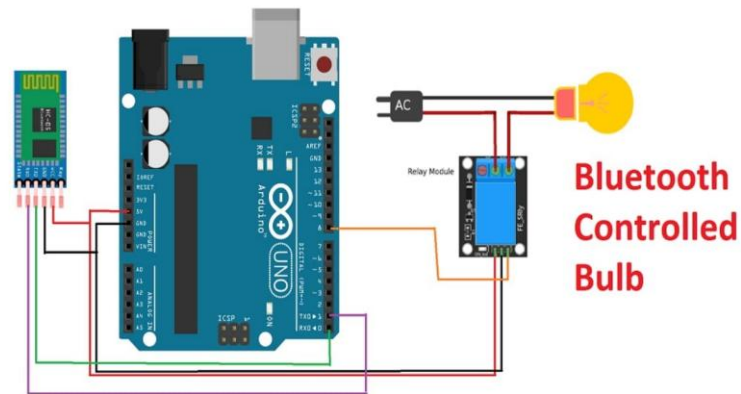


Fig.07.3: Circuit Diagram of Home Automation System using HC-05 and Arduino

Experimental Setup:

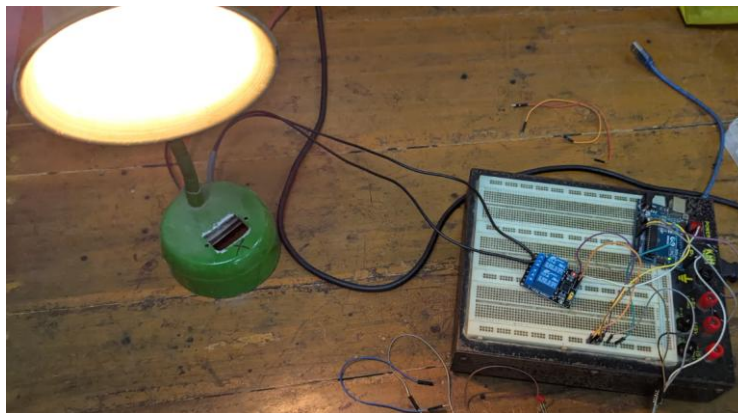


Fig.07.4: Experimental Setup of Home Automation System using HC-05 and Arduino

Code:

<pre>// Include SoftwareSerial for Bluetooth communication #include <SoftwareSerial.h> // Create software serial object for HC-05 SoftwareSerial bluetooth(10, 11); // RX, TX</pre>	<pre>void loop() { // Check if data is available from HC-05 if (bluetooth.available()) { char command = bluetooth.read(); // Read the character sent Serial.print("Received: "); Serial.println(command); } }</pre>
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<pre>// Define relay pins const int relay1 = 2; // Channel 1 of relay module void setup() { // Start serial communication with Bluetooth bluetooth.begin(9600); // Set relay pin as output pinMode(relay1, OUTPUT); // Ensure relay is off initially digitalWrite(relay1, HIGH); // HIGH = OFF for active LOW relays // (Optional) Begin Serial Monitor for debugging Serial.begin(9600); Serial.println("System Ready. Waiting for Bluetooth commands..."); }</pre>	<pre>// Control Relay 1 (bulb) if (command == 'A') { digitalWrite(relay1, LOW); // Turn ON bulb (active LOW) bluetooth.println("Bulb ON"); } else if (command == 'a') { digitalWrite(relay1, HIGH); // Turn OFF bulb bluetooth.println("Bulb OFF"); } }</pre>
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Experimental Procedure:

1. The HC-05 Bluetooth module is connected to the Arduino Uno as follows:
 - o VCC is connected to 5V
 - o GND is connected to GND
 - o TX is connected to Arduino pin 10 (through a voltage divider)
 - o RX is connected to Arduino pin 11
2. The 2-channel relay module is connected to the Arduino Uno:
 - o VCC is connected to 5V
 - o GND is connected to GND
 - o IN1 is connected to Arduino pin 2
3. The AC bulb is wired to the relay module:
 - o One terminal of the bulb is connected to the normally open (NO) pin of relay channel 1
 - o The common (COM) pin of the relay is connected to the live wire of the AC power source
 - o The other terminal of the bulb is connected to the neutral wire
4. The Arduino code is uploaded to the Arduino Uno using the Arduino IDE
5. A Bluetooth terminal application is opened on a smartphone and the HC-05 module is paired.
6. The paired HC-05 device is opened in the app, and control commands are sent:
 - o 'A' is sent to turn ON the bulb
 - o 'a' is sent to turn OFF the bulb
7. The status of the bulb is observed as it is controlled by the Arduino based on the received Bluetooth commands

Hands-on Learning Projects for Mastery of Home Automation using Arduino and HC-05 Bluetooth Module

Task No.: 01

Task Name: Dual Mode Control System for Light Automation (Manual and Auto Switching)

Objective:

- To design and implement a control system that switches between manual and automatic modes for light automation. In manual mode, the light bulb is controlled directly via Bluetooth commands ('1' for ON, '0' for OFF). In automatic mode, a timer or sensor-based logic automatically controls the light's state, ensuring dynamic operation.

Code:

<pre>#include <SoftwareSerial.h> #include <RTCLib.h> // Optional: For Real-Time Clock if using real-time scheduling // SoftwareSerial for HC-05 Bluetooth module SoftwareSerial bluetooth(10, 11); // RX, TX // Relay pin const int relay1 = 2; // Mode flag bool autoMode = false; // false = Manual, true = Auto void setup() { pinMode(relay1, OUTPUT); digitalWrite(relay1, HIGH); // Relay OFF (active LOW) bluetooth.begin(9600); Serial.begin(9600); Serial.println("Dual Mode Light Control System Ready."); Serial.println("Send 'm' for Manual mode, 'a' for Auto mode."); Serial.println("In Manual mode, send '1' to turn ON and '0' to turn OFF."); } void loop() { // Check for Bluetooth input if (bluetooth.available()) { char command = bluetooth.read(); Serial.print("Received: "); Serial.println(command);</pre>	<pre>// Manual Control if (!autoMode) { if (command == '1') { digitalWrite(relay1, LOW); // Turn ON bluetooth.println("Light ON (Manual)"); } else if (command == '0') { digitalWrite(relay1, HIGH); // Turn OFF bluetooth.println("Light OFF (Manual)"); } } // Automatic Control Logic if (autoMode) { int currentHour = hour(); // Simulated time; replace with RTC.hour() if using RTC // Simulate nighttime hours from 18 (6 PM) to 6 (6 AM) if (currentHour >= 18 currentHour < 6) { digitalWrite(relay1, LOW); // Light ON } else { digitalWrite(relay1, HIGH); // Light OFF } } delay(500); // Basic debounce/delay }</pre>
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<pre> if (command == 'm') { autoMode = false; bluetooth.println("Manual Mode Activated"); } else if (command == 'a') { autoMode = true; bluetooth.println("Automatic Mode Activated"); } </pre>	<pre> // Simulated function for current hour (Replace with RTC if needed) int hour() { // Simulate hour using millis() (1 sec = 1 hour simulation for testing) unsigned long simulatedMillis = millis() / 1000; // Fast simulation return simulatedMillis % 24; // Return hour from 0 to 23 } </pre>
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Task No.: 02

Task Name: Voice-Activated Smart Home Control using Mobile Assistant and Bluetooth Communication

Objective:

- To develop a smart home control system that uses mobile assistant voice commands (e.g., Google Assistant) to control a relay module via the HC-05 Bluetooth module. The system interprets the voice commands through Arduino to control a light bulb or motor in an automatic or manual fashion based on the user's spoken input.

Code:

<pre> #include <SoftwareSerial.h> // Bluetooth communication on pins 10 (RX) and 11 (TX) SoftwareSerial bluetooth(10, 11); // RX, TX // Relay pin definition const int relay1 = 2; void setup() { pinMode(relay1, OUTPUT); digitalWrite(relay1, HIGH); // OFF initially (active LOW relay) bluetooth.begin(9600); Serial.begin(9600); Serial.println("Voice-Activated Smart Home Control Ready."); } void loop() { // Check for incoming Bluetooth data if (bluetooth.available()) { </pre>	<pre> } else if (command.equalsIgnoreCase("turn off light") command.equalsIgnoreCase("light off")) { digitalWrite(relay1, HIGH); // Turn OFF relay bluetooth.println("Light turned OFF"); } else if (command.equalsIgnoreCase("turn on motor")) { digitalWrite(relay1, LOW); // You can use same relay or separate one bluetooth.println("Motor turned ON"); } else if (command.equalsIgnoreCase("turn off motor")) { digitalWrite(relay1, HIGH); bluetooth.println("Motor turned OFF"); </pre>
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<pre>String command = bluetooth.readStringUntil('\n'); // Read full line command.trim(); // Remove any trailing newline or spaces Serial.print("Received Command: "); Serial.println(command); // Match against expected voice-to- text command strings if (command.equalsIgnoreCase("turn on light") command.equalsIgnoreCase("light on")) { digitalWrite(relay1, LOW); // Turn ON relay (bulb/motor) bluetooth.println("Light turned ON");</pre>	<pre> } else { bluetooth.println("Unrecognized command"); } }</pre>
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Discussions:

The experiments were successfully done using bluetooth and relay module with arduino.

Justification of CO2 and PO(j):

The designs enable safe, timed, and condition-based control of a light bulb through manual, automatic, and voice commands. They showcase effective use of Bluetooth, relay control, and embedded logic for smart home automation.

Conclusion:

These projects demonstrate how embedded systems can automate home automation control. It highlights the potential of combining Bluetooth, sensors, and microcontrollers for smart home automation.

Reference:

1. <https://www.geeksforgeeks.org/all-about-hc-05-bluetooth-module-connection-with-android/>
2. <https://osoyoo.com/2017/07/31/2-channel-relay-module-2/>